

Downside Performance Harness v0.1

SOTA-grade downside-adjusted hypothetical backtested performance under the included deterministic downside_performance_v1 benchmark suite.

Scoped Claim

This paper documents a bounded performance claim for a research harness, not a live trading system. The claim is limited to deterministic synthetic OHLCV data, included baselines, included robustness checks, and the official performance claim gate.

Field	Value
Claim id	downside_adjusted_backtested_performance_v0_1
Benchmark suite	downside_performance_v1
Performance type	hypothetical_backtested_performance
Gate status	passed / performance_claim_publishable=True

Headline Result

Metric	agentic_candidate_v1
Return multiple	1.872x
Total return	87.25%
CAGR	23.29%
Max drawdown	0.79%
Calmar	29.49
Sharpe	14.44
Average exposure	68.25%

Interpretation: the candidate is top-ranked on total return, CAGR, and Calmar within the included benchmark, while preserving the non-claims required by the claim contract.

Benchmark Method

The benchmark executes a fixed deterministic strategy registry: cash, buy-and-hold synthetic SPY, equal weight, SMA crossover, simple momentum, mean reversion, volatility targeting, the Stock Harness MA-to-cash baseline, and agentic_candidate_v1.

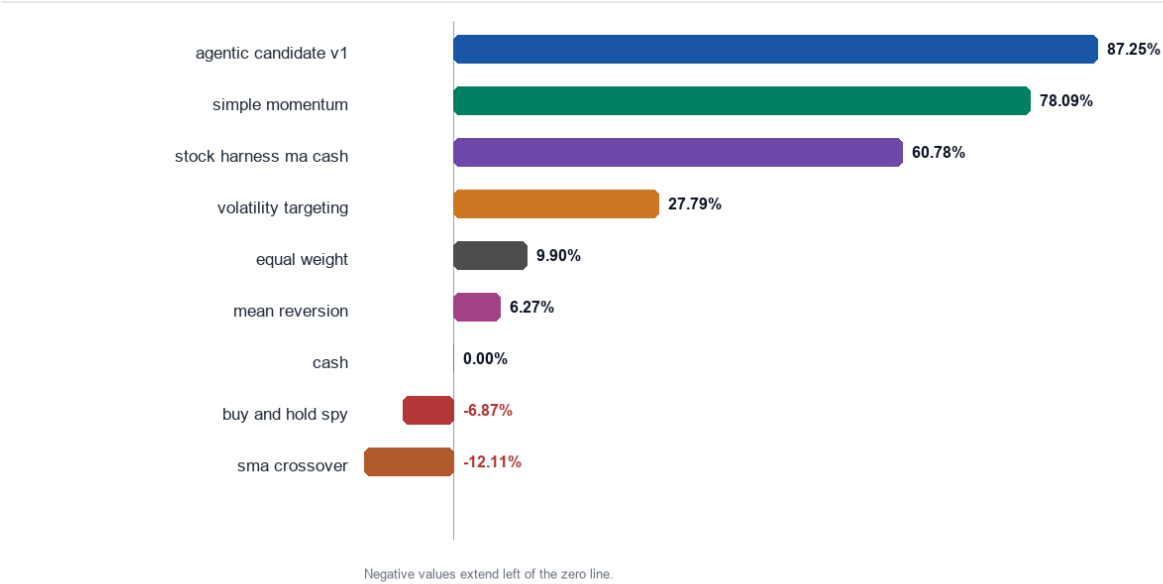
Measured metrics include total return, return multiple, CAGR, annualized return, max drawdown, volatility, downside deviation, Sharpe, Sortino, Calmar, worst month, worst year, turnover, exposure, and final equity.

Baseline Table

Rank	Strategy	Total Return	CAGR	Max DD	Calmar	Sharpe
1	agentic_candidate_v1	87.25%	23.29%	0.79%	29.49	14.44
2	simple_momentum	78.09%	21.24%	1.81%	11.72	13.69
3	stock_harness_ma_cash	60.78%	17.17%	0.65%	26.42	14.53
4	volatility_targeting	27.79%	8.53%	12.83%	0.66	2.80
5	equal_weight	9.90%	3.20%	16.40%	0.20	1.10
6	mean_reversion	6.27%	2.05%	10.88%	0.19	0.69
7	cash	0.00%	0.00%	0.00%	0.00	0.00
8	buy_and_hold_spy	-6.87%	-2.35%	30.69%	-0.08	-0.42
9	sma_crossover	-12.11%	-4.22%	22.75%	-0.19	-0.83

Total Return by Strategy

Included deterministic baselines plus agentic_candidate_v1.

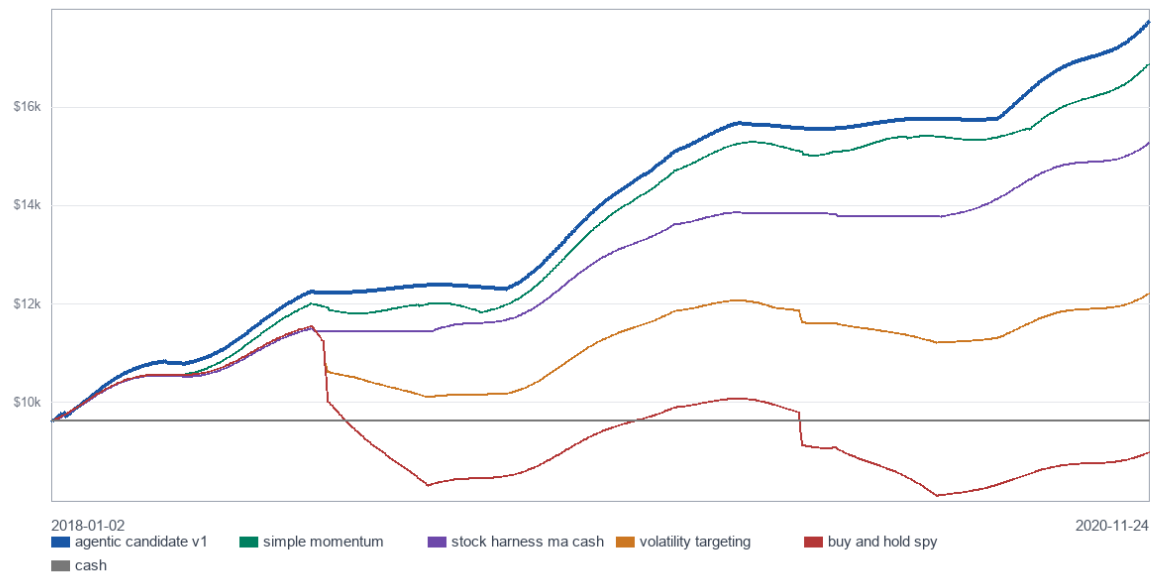


Equity and Drawdown

The equity curve and drawdown curve show the candidate against the strongest included baselines. These figures are hypothetical backtested curves generated from the same deterministic evidence packet used by the claim gate.

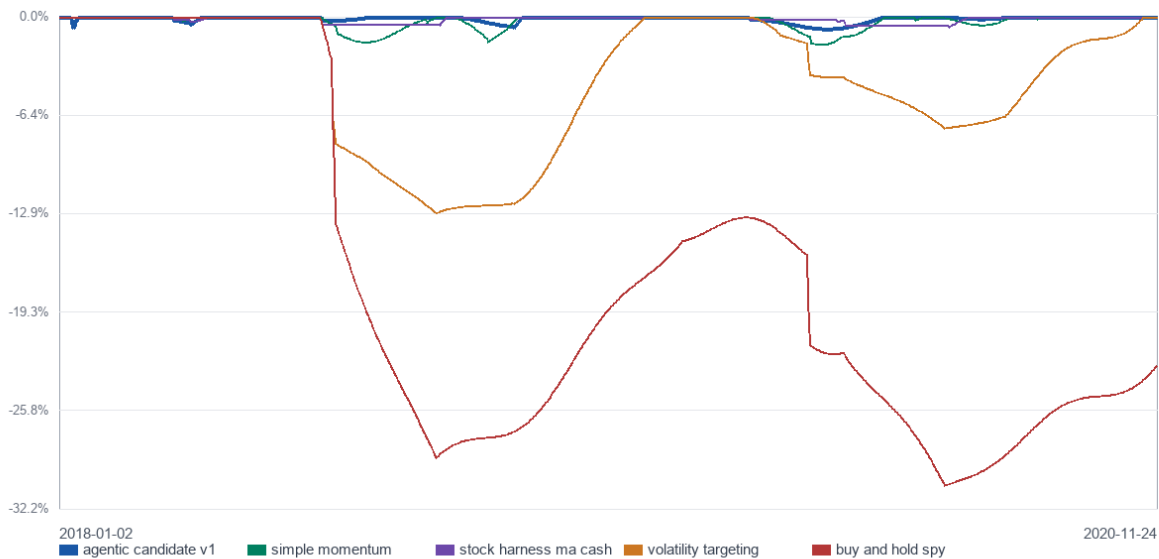
Equity Curve Comparison

Hypothetical backtested equity on the included downside_performance_v1 benchmark.



Drawdown Curve Comparison

Lower magnitude drawdowns are better; values are shown as negative percentages.

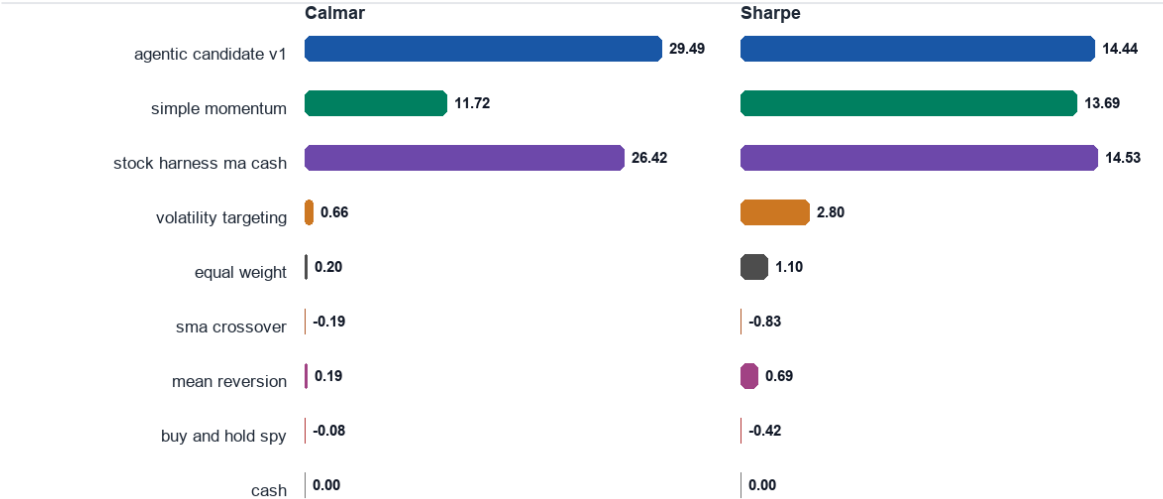


Risk-Adjusted Ranking

The Calmar ratio captures return per unit of max drawdown. In this benchmark, agentic_candidate_v1 is ranked first on Calmar and remains competitive on Sharpe while using downside-aware exposure control.

Risk-Adjusted Ranking

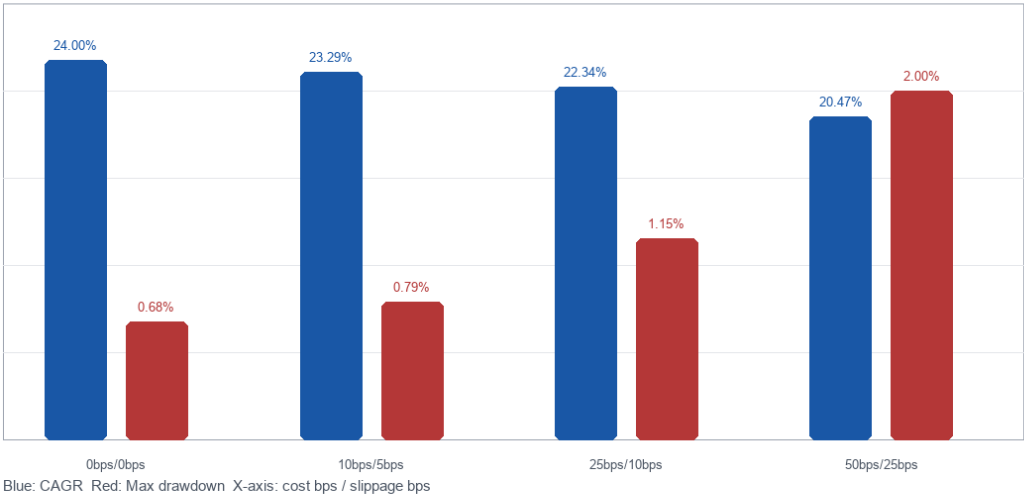
Calmar emphasizes return per unit of max drawdown; Sharpe uses annualized volatility.



Cost and Slippage Stress

Cost and Slippage Stress

agentic_candidate_v1 remains publishable under the included stress ladder.



Robustness Checks

Check	Result
Data quality	True
Lookahead audit	True
Walk-forward	True
Cost/slippage stress	True
Parameter sensitivity	True
Negative controls	True

Defense Layer v0.2

The public snapshot includes a defense packet for reviewer-facing attack surfaces: strategy freeze, data lineage and bias disclosure, baseline fairness, deterministic bootstrap confidence intervals, and a forward paper-trading protocol. This layer does not expand the claim beyond hypothetical backtested performance under the included benchmark.

Defense check	Passed
Strategy freeze	True
Data lineage / bias	True
Baseline fairness	True
Bootstrap confidence	True
Forward paper-trading protocol	True
Claim boundary preserved	True
Defense gate	passed / True

Bootstrap metric	p05 / median / p95
CAGR	16.85% / 23.34% / 29.61%
Max drawdown	0.45% / 0.76% / 0.93%
Calmar ratio	20.54 / 32.11 / 57.13
Sharpe ratio	11.49 / 14.60 / 18.64

Forward paper-trading protocol starts on 2026-05-27 with checkpoints 2026-06-26, 2026-08-25, 2026-11-23. These checkpoints are for future evidence collection only and are not live-performance claims.

Non-Claims

- No financial advice.
- No live trading readiness claim.
- No return guarantee or future performance claim.
- No realized investor return claim.
- No broker integration, order routing, or execution-readiness claim.
- No universal market, strategy, or external-framework dominance claim.
- No claim outside the included downside_performance_v1 benchmark suite.

Evidence Packet

The public snapshot packages the claim-gate JSON, benchmark JSON, evidence packet, charts, Markdown paper, and PDF paper with SHA-256 hashes. The official result is only publishable when the deterministic claim gate reports performance_claim_publishable=true.